

Q1 2021 (01 Sep Shift 2)

A uniform heating wire of resistance 36Ω is connected across a potential difference of 240 V. The wire is then cut into half and potential difference of 240 V is applied across each half separately. The ratio of power dissipation in first case to the total power dissipation in the second case would be 1 : x, where x is.....

Q2 2021 (01 Sep Shift 2)

Two resistors $R_1 = (4 \pm 0.8)\Omega$ and $R_2 = (4 \pm 0.4)\Omega$ are connected in parallel. The equivalent resistance of their parallel combination will be :

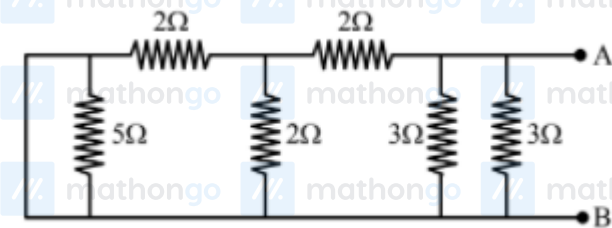
- (1) $(4 \pm 0.4)\Omega$
- (2) $(2 \pm 0.4)\Omega$
- (3) $(2 \pm 0.3)\Omega$
- (4) $(4 \pm 0.3)\Omega$

Q3 2021 (31 Aug Shift 2)

A resistor dissipates 192 J of energy in 1 s when a current of 4 A is passed through it. Now, when the current is doubled, the amount of thermal energy dissipated in 5 s is..... J.

Q4 2021 (31 Aug Shift 2)

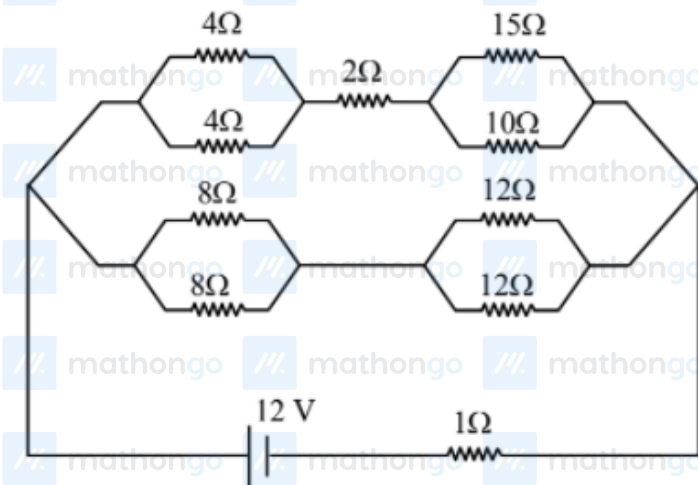
The equivalent resistance of the given circuit between the terminals A and B is :



- (1) 0Ω
- (2) 3Ω
- (3) $\frac{9}{2}\Omega$
- (4) 1Ω

Q5 2021 (31 Aug Shift 1)

The voltage drop across 15Ω resistance in the given figure will be V.



Q6 2021 (31 Aug Shift 1)

A square shaped wire with resistance of each side 3Ω is bent to form a complete circle. The resistance between two diametrically opposite points of the circle in unit of Ω will be

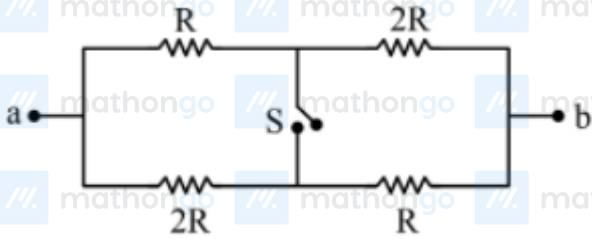
Q7 2021 (31 Aug Shift 1)

Consider a galvanometer shunted with 5Ω resistance and 2% of current passes through it. What is the resistance of the given galvanometer?

- (1) 300Ω
- (2) 344Ω
- (3) 245Ω
- (4) 226Ω

Q8 2021 (27 Aug Shift 2)

The ratio of the equivalent resistance of the network (shown in figure) between the points a and b when switch is open and switch is closed is $x : 8$. The value of x is



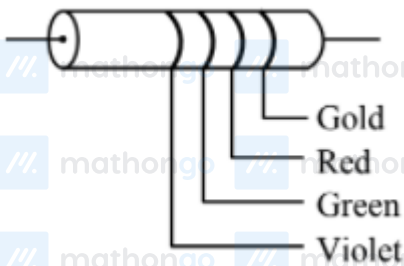
Q9 2021 (27 Aug Shift 2)

For full scale deflection of total 50 divisions, 50mV voltage is required in galvanometer. The resistance of galvanometer if its current sensitivity is 2 div/mA will be :

- (1) 1Ω
- (2) 5Ω
- (3) 4Ω
- (4) 2Ω

Q10 2021 (27 Aug Shift 2)

The colour coding on a carbon resistor is shown in the given figure. The resistance value of the given resistor is :



- (1) $(5700 \pm 285)\Omega$
- (2) $(7500 \pm 750)\Omega$
- (3) $(5700 \pm 375)\Omega$
- (4) $(7500 \pm 375)\Omega$

Q11 2021 (27 Aug Shift 1)

First, a set of n equal resistors of 10Ω each are connected in series to a battery of emf 20 V and internal resistance 10Ω . A current I is observed to flow. Then, the n resistors are connected in parallel to the same battery. It is observed that the current is increased 20 times, then the value of n is .

Q12 2021 (27 Aug Shift 1)

Five identical cells each of internal resistance 1Ω and emf 5 V are connected in series and in parallel with an external resistance 'R'. For what value of 'R', current in series and parallel combination will remain the same?

- (1) 1Ω
- (2) 25Ω
- (3) 5Ω
- (4) 10Ω

Q13 2021 (26 Aug Shift 2)

An electric bulb of 500 watt at 100 volt is used in a circuit having a 200 V supply. Calculate the resistance R to be connected in series with the bulb so that the power delivered by the bulb is 500 W .

- (1) 20Ω
- (2) 30Ω
- (3) 5Ω
- (4) 10Ω

Q14 2021 (26 Aug Shift 2)

If you are provided a set of resistances 2Ω , 4Ω , 6Ω and 8Ω . Connect these resistances so as to obtain an equivalent resistance of $\frac{46}{3}\Omega$

- (1) 4Ω and 6Ω are in parallel with 2Ω and 8Ω in series
- (2) 6Ω and 8Ω are in parallel with 2Ω and 4Ω in series
- (3) 2Ω and 6Ω are in parallel with 4Ω and 8Ω in series
- (4) 2Ω and 4Ω are in parallel with 6Ω and 8Ω in series

Q15 2021 (26 Aug Shift 1)

What equal length of an iron wire and a copper-nickel alloy wire, each of 2 mm diameter connected parallel to give an equivalent resistance of 3Ω ?

(Given resistivities of iron and copper-nickel alloy wire are $12\mu\Omega\text{cm}$ and $51\mu\Omega\text{cm}$ respectively)

(1) 82m

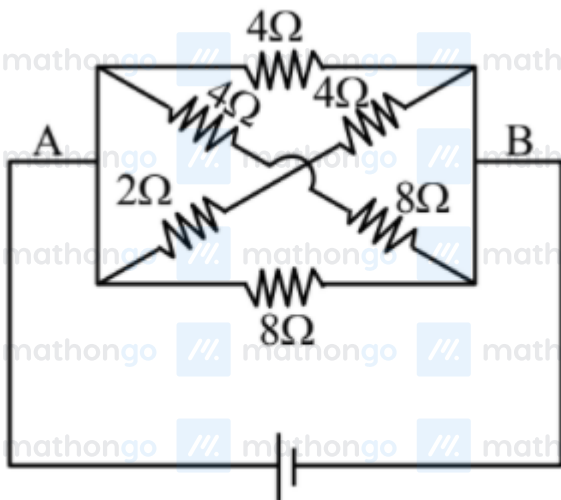
(2) 97m

(3) 110m

(4) 90m

Q16 2021 (26 Aug Shift 1)

In the given figure, the emf of the cell is 2.2 V and if internal resistance is 0.6Ω . Calculate the power dissipated in the whole circuit:



2.2 V, $r=0.6\Omega$

(1) 1.32W

(2) 0.65W

(3) 2.2W

(4) 4.4W

Answer Key

Q1 (4)

Q2 (3)

Q3 (3840)

Q4 (4)

Q5 (6)

Q6 (3)

Q7 (3)

Q8 (9)

Q9 (4)

Q10 (4)

Q11 (20)

Q12 (1)

Q13 (1)

Q14 (4)

Q15 (2)

Q16 (3)

Q1 (4)

$$\text{First case } P_1 = \frac{V^2}{R} = \frac{(240)^2}{36}$$

Second case Resistance of each half = 18Ω

$$P_2 = \frac{(240)^2}{18} + \frac{(240)^2}{18} = \frac{(240)^2}{9}$$

$$\frac{P_1}{P_2} = \frac{1}{4}$$

$$x = 4.00$$

Q2 (3)

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_{eq}} = \frac{1}{4} + \frac{1}{4} \Rightarrow R_{eq} = 2\Omega$$

$$\text{Also } \frac{\Delta R_{eq}}{R_{eq}^2} = \frac{\Delta R_1}{R_1^2} + \frac{\Delta R_2}{R_2^2}$$

$$\frac{\Delta R_{eq}}{4} = \frac{.8}{16} + \frac{.4}{16} = \frac{1.2}{16}$$

$$\Delta R_{eq} = 0.3\Omega$$

$$R_{eq} = (2 \pm 0.3)\Omega$$

Option (3)

Q3 (3840)

$$E = i^2 R t$$

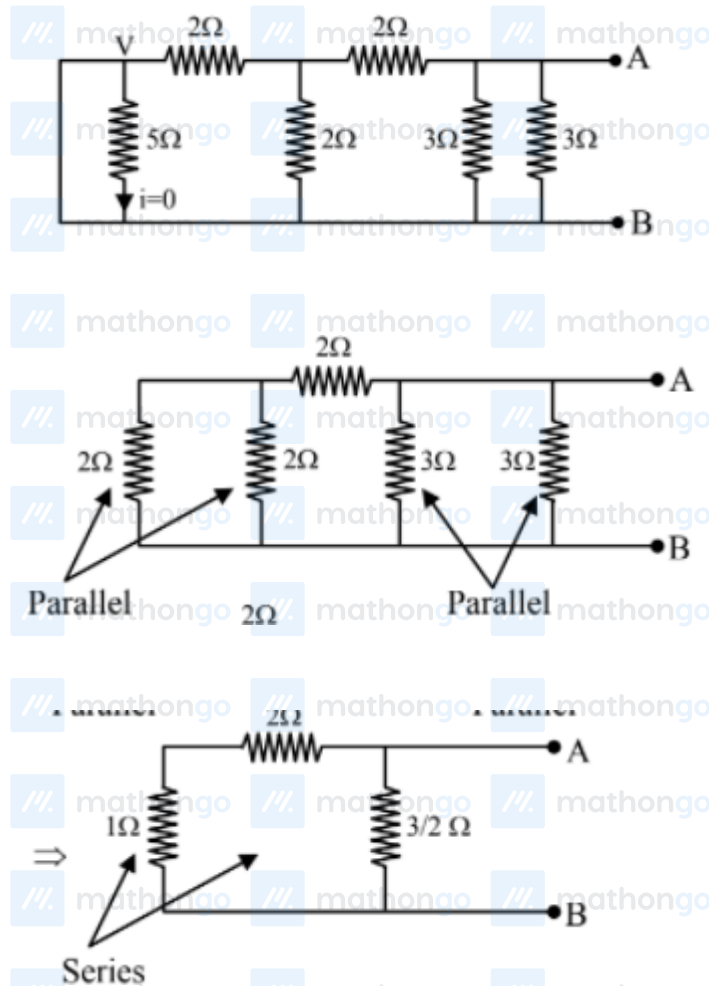
$$192 = 16(R)(1)$$

$$R = 12\Omega$$

$$E^1 = (8)^2(12)(5)$$

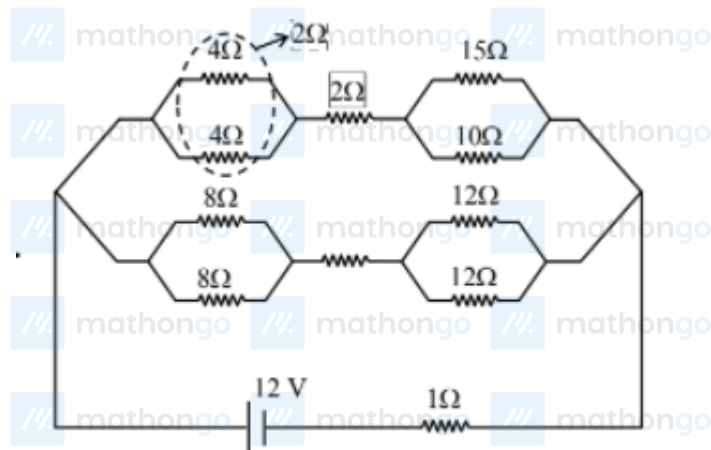
$$= 3840 \text{ J}$$

Q4 (4)

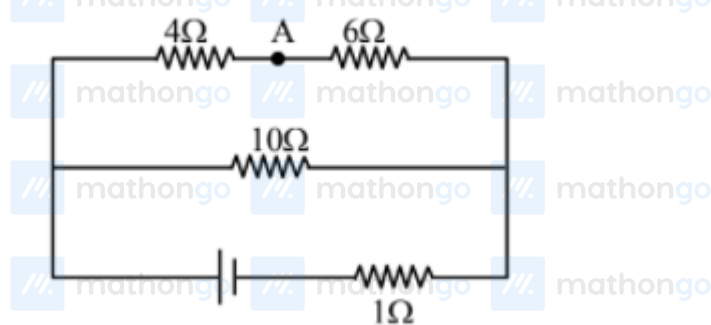
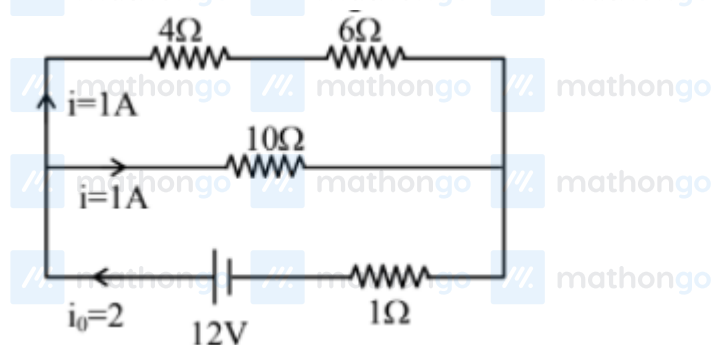


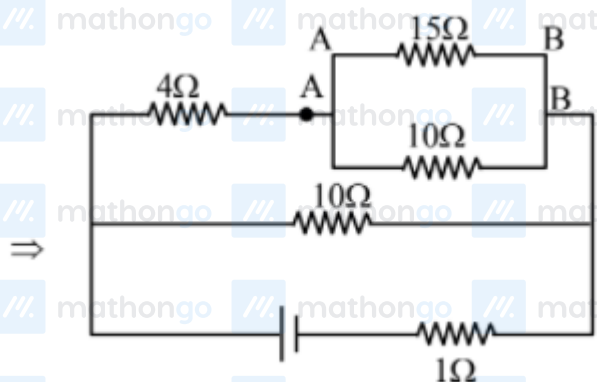
$$R_{eq} = \frac{3 \times 3/2}{3 + 3/2} = \frac{9/2}{9/2} = 1\Omega$$

Q5 (6)



⇒ effective circuit diagram will be

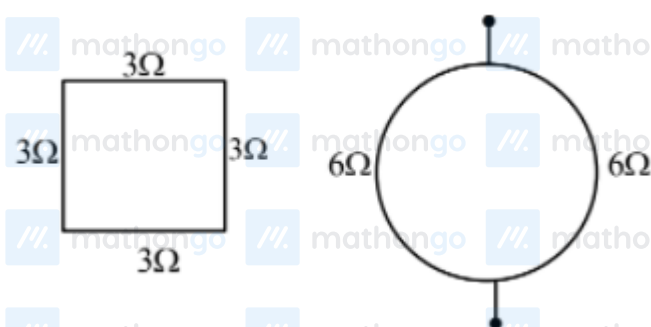




Point drop across $6\Omega = 1 \times 6 = 6 = V_{AB}$

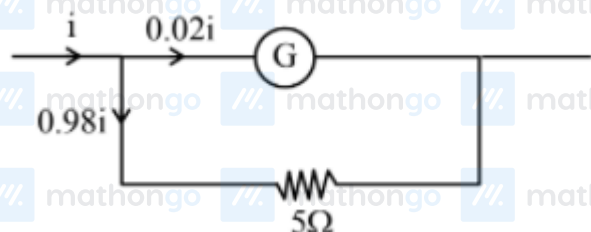
$\Rightarrow$ Hence point drop across $15\Omega = 6 \text{ volt} = V_{AB}$

Q6 (3)



$R_{cq} = 3\Omega$

Q7 (3)



$$0.02iR_g = 0.98i \times 5$$

$$R_g = 245\Omega$$

Option (3)

Q8 (9)

$$R_{\text{eq open}} = \frac{3R}{2}$$

$$R_{\text{eq closed}} = 2 \times \frac{R \times 2R}{3R} = \frac{4R}{3}$$

$$\therefore R_{\text{eq open}} R_{\text{eq closed}} = \frac{3R}{2} \times \frac{4R}{3} = \frac{9}{8}$$

$$\therefore x = 9$$

Q9 (4)

$$I_{\text{max}} = \frac{50}{2} = 25 \text{ mA}$$

$$R = \frac{V}{I} = \frac{50 \text{ mV}}{25 \text{ mA}} = 2\Omega$$

Q10 (4)

$$R = 75 \times 10^2 \pm 5\% \text{ of } 7500$$

$$R = (7500 \pm 375)\Omega$$

Q11 (20)

In series

$$R_{\text{eq}} = nR = 10n$$

$$i_s = \frac{20}{10+10n} = \frac{2}{1+n}$$

in parallel

$$R_{eq} = \frac{10}{n}$$

$$i_p = \frac{20}{\frac{10}{n} + 10} = \frac{2n}{1+n}$$

$$\frac{i_p}{i_s} = 20$$

$$\left(\frac{2n}{1+n} \right) = 20$$

$$\left(\frac{2}{1+n} \right) 20 = 20$$

$$n = 20$$

Q12 (1)

$$i_1 = \frac{25}{5+R}$$

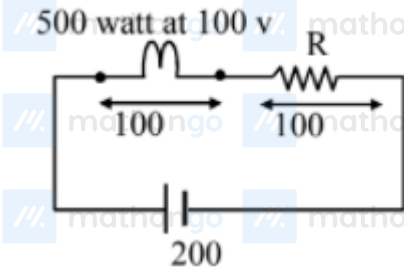
$$i_2 = \frac{5}{R + \frac{1}{5}}$$

$$i_1 = i_2 \Rightarrow 5 \left(R + \frac{1}{5} \right) = 5 + R$$

$$4R = 4$$

$$R = 1\Omega$$

Q13 (1)



$$P = Vi$$

$$500 = Vi$$

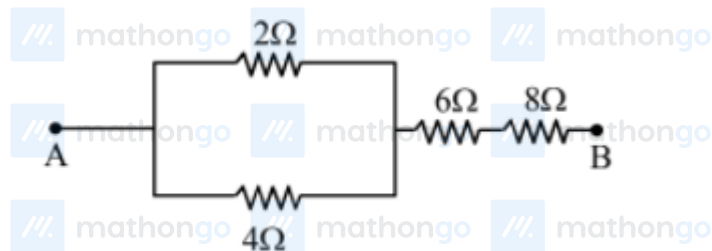
$$i = 5\text{Amp}$$

$$V = i \times R$$

$$R = 20$$

Ans. (1)

Q14 (4)



$$\frac{46}{3} \Omega$$

Q15 (2)

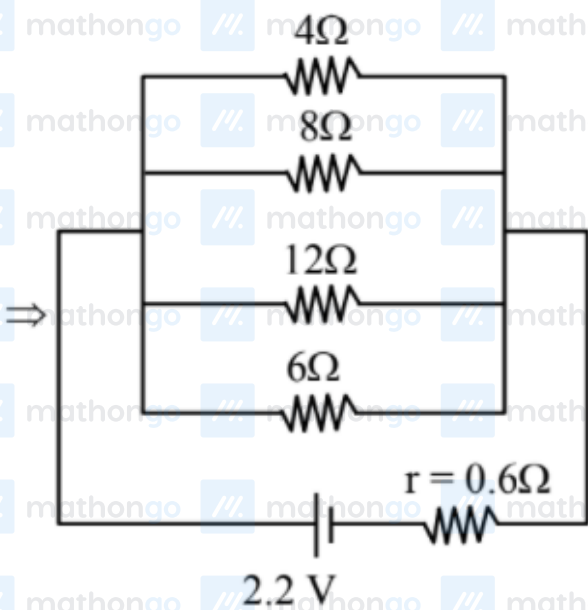
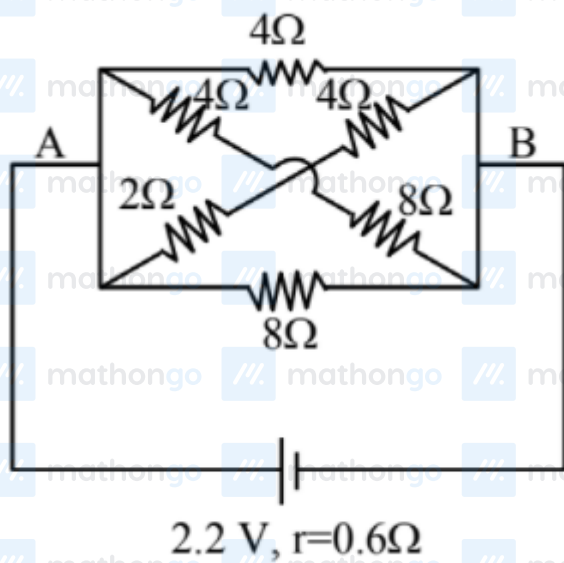
$$\frac{R_1 R_2}{R_1 + R_2} = 3$$

$$\frac{(12 \times 10^{-6} \times 10^{-2}) \ell \times 4}{\pi (2)^2 \times 10^{-6}} \times \frac{(51 \times 10^{-6} \times 10^{-2}) \ell \times 4}{\pi (2)^2 \times 10^{-6}} = \frac{63 \times 10^{-6} \times 10^{-2} \times \ell \times 4}{\pi (2)^2 \times 10^{-6}}$$

$$\Rightarrow \ell = 97 \text{ m}$$

Option (2)

Q16 (3)



$$\frac{1}{R_{eq}} = \frac{1}{4} + \frac{1}{8} + \frac{1}{12} + \frac{1}{6} = \frac{6+3+2+4}{24} = \frac{15}{24}$$

$$R_{eq} = \frac{24}{15} = 1.6 \Rightarrow R_T = 1.6 + 0.6 = 2.2\Omega$$

$$P = \frac{V^2}{R_T} = \frac{(2.2)^2}{2.2} = 2.2 \text{ W}$$

#MathBoleTohMathonGo